The summand-swap invariant does not refute below degree 4 at ℓ = 6; degree 4 at ℓ = 7 is unresolved

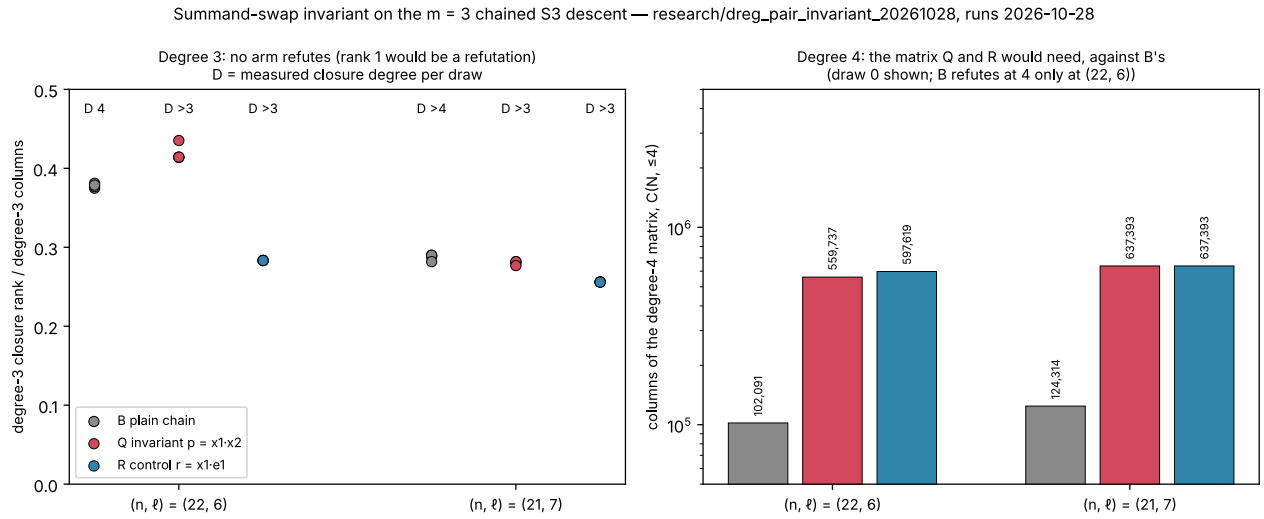
Study `research/dreg_pair_invariant_20261028/`, pre-registered. **Class:** stage diagnostic. **Verdict:** H_none in both cells. At (22, 6) that is measured. At (21, 7) it follows the registered convention for an unmeasured degree 4, and that cell is unresolved.

Question

At m = 3 every cubic monomial of the chained descent comes from $x_1x_2e_1$. In the swap invariants $s = x_1+x_2$ and $p = x_1x_2$, the first link becomes the quadratic $s^2e_1^2 + p\,e_1 + p^2 + b$.

Does adding p as unknowns let the MutantXL closure refute at a lower degree? Is any gain due to the invariant, or only to quadratisation? Quadratisation alone is tested by a matched control, $r = x_1e_1$.

Result



cell	B (plain)	Q (invariant)	R (control)
(22, 6), 4 draws	refutes at 4	> 3	> 3
(21, 7), 4 draws	> 4	> 3	> 3

- Q's degree 4 is unmeasured. Its dense matrix needs 523,686–637,393 columns, against B's 102,091 and 124,314.
- So at (22, 6) the invariant arm refutes no earlier than B, at 5.1–5.5 times the columns.
- At (21, 7), B does not refute at 4 (C(42, ≤ 5) = 974,982 columns at degree 5). If Q refuted at 4, with 637,393 columns, it would be lower and smaller. That is not measured, so this cell is open.
- At degree 3, quadratisation alone (R) does not help either.
- Controls valid: exact identity check, B reproduces its records, and no satisfiable draw is refuted.

Reading

- **Where the plain chain refutes at 4 (ℓ = 6), the pair invariant does not help** (measured).
- **Where it does not refute at 4 (ℓ = 7), it is unresolved:** degree 4 of Q at (21, 7) decides it and needs a large-memory host.
- At m = 4 the second cubic link has no summand symmetry.
- **Unregistered pointer:** at (22, 6), Q closes more of its degree-3 matrix (0.41–0.44) than B (0.37–0.38) or R (0.28), but not enough to refute.
- **No row moves and none beats rho.** There is no improvement over the baseline yet.
- **Untried successors:** the invariants as a replacement for the summand coordinates, so that N falls; the orbit-aware split rule.